

AFAT UDDIN

Software Engineer | Generative AI

Dhaka, Bangladesh | +8801851858804 | afatuddinshaon@gmail.com

[LinkedIn](#) | [GitHub](#) | [YouTube](#)

PROFESSIONAL SUMMARY

Software Engineer with 3 years of experience designing and shipping production-grade backend systems in Python and Django, with a current focus on Generative AI applications. Proven track record building multi-tenant SaaS platforms that integrate LLMs (Google Gemini, Replicate), RAG pipelines (LangChain + pgvector), and async task workflows (Celery + Redis) at scale. Skilled in REST API design, cloud deployment on Azure and AWS, payment integrations (Stripe), and CI/CD automation. Passionate about turning AI capabilities into reliable, well-tested, and maintainable products.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, SQL

Backend Frameworks: Django, Django REST Framework, FastAPI, Flask

Generative AI & ML: Google Gemini API, Replicate API, LangChain, RAG, pgvector, BGE Embeddings, Prompt Engineering, OpenAI API

Async & Messaging: Celery, Celery Beat, Redis

Databases: PostgreSQL (pgvector), MySQL, MongoDB, SQLite, Redis

Cloud & DevOps: Azure (Blob Storage, Container Apps, Pipelines, DevOps), AWS (SES, RDS), Docker, Nginx, Supervisor, Gunicorn, Cloudinary

Observability: Prometheus, Grafana, Fluentd, Sentry

APIs & Integrations: Stripe, Google OAuth2, Meta/LinkedIn/Twitter Ads APIs, SendGrid, Mailgun, Pexels, Unsplash, Shutterstock

Web Scraping & Automation: BeautifulSoup, Scrapy, Selenium, Requests

Testing & Documentation: Pytest, factory_boy, pytest-django, drf-spectacular (OpenAPI/Swagger)

Frontend (Working Knowledge): React.js, HTML, CSS, Tailwind CSS, Bootstrap

Tools & Practices: Git, GitHub, Azure DevOps, Linux, REST API Design, JWT Auth, OAuth2, Multi-tenant SaaS, CI/CD

Spoken Languages: English (Fluent)

PROFESSIONAL EXPERIENCE

Backend Developer

CodefusionAI

Jan 2025 – Present

Dhaka, Bangladesh

- Lead backend development of two production Generative AI SaaS platforms — CreativeGenie (AI content creation) and Creative Experience (AI-powered document analytics) — built on Django + DRF with multi-tenant workspaces and role-based access control.
- Engineered LLM integrations with Google Gemini and Replicate model APIs behind a clean service layer, powering AI image generation, text generation, magic eraser/replace, image upscaling, retouching, and AI-generated document descriptions.
- Built RAG and semantic search using LangChain + pgvector with RunPod-hosted BGE embeddings, enabling vector search inside the existing PostgreSQL instance and reducing inference cost.
- Designed async document- and asset-processing pipelines on Celery + Redis (with Celery Beat) and optimized upload-to-ready time by 80% for heavy AI workloads.

- Implemented Stripe-based subscription billing (plans, checkout, webhooks, trial assignment) and custom DRF permission classes (IsValidSubscription, IsUserInWorkspace) to gate premium features.
- Shipped ads-platform integrations for Meta, LinkedIn, and Twitter — OAuth account linking, ad-account eligibility checks, campaign creation, and automated status synchronization.
- Authored CI/CD on Azure DevOps across dev / staging / prod, multi-stage Docker builds for web and worker services orchestrated with Supervisor (Gunicorn + Nginx), and Sentry observability.
- Set up reliability and quality controls: DRF throttling, a normalized API error envelope, drf-spectacular OpenAPI docs, and a pytest + factory_boy + pytest-django test suite wired into CI.

Junior Software Engineer (Python, Django)

Jan 2023 – Dec 2024

Mediusware Ltd.

Dhaka, Bangladesh

- Developed and maintained robust web applications using Python, Django, and Django REST Framework, contributing to API design and implementation for scalable backend systems.
- Built a SaaS Review Management System with Google OAuth2 sign-in and Google Business API integration to retrieve location reviews, plus multi-channel outreach (Email, SMS, WhatsApp) for review requests.
- Integrated third-party APIs to monitor and verify customer reviews across relevant platforms, with Celery + Redis handling async delivery.
- Implemented web scraping and automation pipelines using BeautifulSoup, Scrapy, Selenium, and Requests to streamline data collection and processing tasks across multiple data sources.
- Containerized services with Docker for consistent local development, staging, and production parity.
- Collaborated with cross-functional teams (frontend, design, QA, product) to deliver efficient and reliable software solutions.

KEY PROJECTS

CreativeGenie — AI Content Creation SaaS Platform

Django 4.2, DRF, PostgreSQL, Celery, Redis, Azure, Stripe, Google Gemini, Replicate

- Engineered a multi-tenant SaaS backend powering an AI content-creation platform with workspace-scoped access control, role-based permissions (admin / client / creative / media), and Google OAuth2 via django-allauth.
- Built 20+ modular Django apps covering AI image generation, text generation, storytelling video, background removal/replacement, magic eraser, magic replace, image upscaling, product photoshoot, and retouching — integrating Google Gemini and Replicate APIs behind a clean service layer.
- Designed and shipped Meta, LinkedIn, and Twitter ads integrations with OAuth linking, ad-account eligibility checks, campaign creation, and automated status synchronization.
- Implemented async task pipelines with Celery 5.4 + Redis (Celery Beat) for image generation, campaign optimization, and asset processing — eliminating request-time latency on heavy AI workloads.
- Integrated Stripe subscription billing with plan seeding, trial assignment, and cancel-at-period-end sync, plus custom DRF permission classes (IsValidSubscription, IsUserInWorkspace) to gate premium features.
- Built reusable abstractions (BaseModel, BaseModelViewSet) to enforce tenant isolation and workspace access on every CRUD operation across the API surface.
- Connected stock-media providers (Pexels, Unsplash, Shutterstock) and an internal asset library with Azure Blob Storage via django-storages for scalable media delivery.
- Generated OpenAPI docs with drf-spectacular (Swagger + ReDoc) and authored a pytest + factory_boy test suite with auto-mocked Sentry and Azure Blob for fast, hermetic CI runs.
- Productionized with multi-stage Docker builds (Nginx + Supervisor), Azure Pipelines CI/CD across dev / staging / prod, and Sentry observability.

Creative Experience — AI-Powered Document Analytics Platform

Django 5.2, DRF, Celery, PostgreSQL/pgvector, Redis, Azure, LangChain, Google Gemini, Stripe

- Built a multi-tenant document-experience backend in Django 5 + DRF with JWT, Google OAuth, and django-allauth, exposing an OpenAPI-documented REST API consumed by a React frontend.
- Designed an async document-processing pipeline on Celery + Redis that orchestrates preview-image extraction, page rendering, TOC parsing, AI description, and vector embedding as a coordinated task chain with select_for_update locking and per-stage status tracking.
- Implemented RAG and semantic search by integrating LangChain with pgvector and RunPod-hosted BGE embeddings — enabling vector search directly inside PostgreSQL while keeping inference cost low.
- Integrated Google Gemini for table-of-contents extraction and AI-generated descriptions, with a PyMuPDF fast path and Gemini fallback so deterministic PDFs skip the LLM entirely.
- Modeled a rich document analytics layer — page-level views, dwell time, device and geo breakdowns, click and heatmap interactions, and per-user / aggregate roll-ups — backed by ~30 related models and 50+ migrations.
- Built a Stripe subscription and billing flow (plans, checkout, webhooks) with seedable plan data via Django management commands.
- Customized Django auth with a UUID-keyed, email-login User model and reusable abstract mixins (TimestampMixin, UserStampedMixin) used across every app.
- Authored CI/CD on Azure DevOps and multi-stage Dockerfiles orchestrated with Supervisor (Gunicorn + Nginx; worker + beat + Nginx sidecar for Azure Container Apps health checks).

Review Management System

Django, DRF, PostgreSQL, Celery, Redis, Mailgun, Docker

- Built a SaaS review-management platform with Google OAuth2 sign-in and Google Business API integration to retrieve location reviews.
- Designed a multi-channel outreach system to send review requests via Email, SMS, and WhatsApp.
- Integrated third-party APIs to monitor and verify customer reviews across relevant platforms.
- Productionized with Celery + Redis for async delivery and Docker for local development parity.

EDUCATION

B.Sc. in Computer Science and Engineering

Ongoing

Prime University, Dhaka, Bangladesh

Diploma in Computer Science and Engineering

Feni Govt. Computer Institute, Feni, Bangladesh

CERTIFICATIONS

- CSE Fundamentals — Phitron
- Web Development — Programming Hero
- Problem Solving (Python, OOP) — HackerRank